LABS 1–11: Probability and Statistics Lab

Diabetes and lifestyle indicators

Statistical analysis with R
BMAT202P Probability and Statistics Lab

YASH RAJ - 24BDS1209
HRISHIKESH DAMODAR NAYAK - 24BDS1007
HARDIK LALLA - 24BDS1010
ARYAN RAJ - 24BDS1094

Data → graphs → models → statistical tests

Dataset and aim

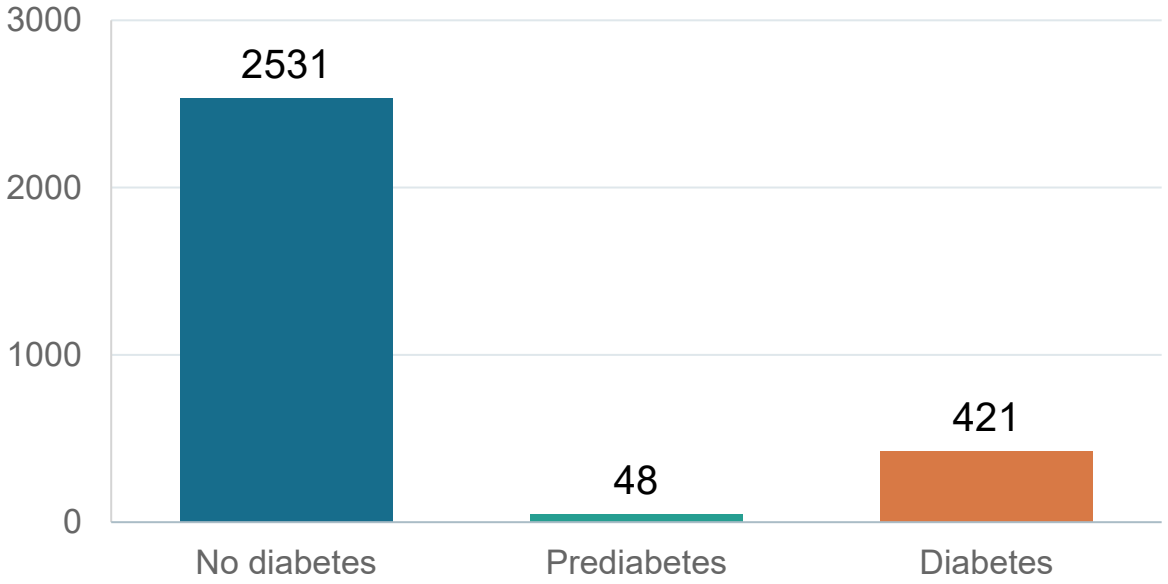
Aim: describe health patterns, study BMI relationships and apply the course statistical methods.

253,680
source records

3,000
sampled people

21
health predictors

Health status in the sample



Key variables

Variable	What it measures
BMI	Body mass index (kg/m²)
Mental / physical health	Poor-health days, 0–30
Age / income / education	Grouped categories
Activity / smoking	Yes / No indicators

Source: CDC BRFSS 2015, processed by Alex Teboul on Kaggle.

469 people (15.63%) report prediabetes or diabetes.

Data preparation

Import → sample → validate → prepare for analysis

Import
CSV source

Sample
3,000 random rows

Validate
Codes and ranges

Prepare
Labels and types

Types used in R

Type	Examples
Numeric	BMI and poor-health days
Categorical	Health status, sex and smoking
Grouped categories	Age, income and education

Quality checks

Check	Result
Missing entries	0
Invalid codes / ranges	0
Health-day range	0–30 days
Sample selection	Seed 101

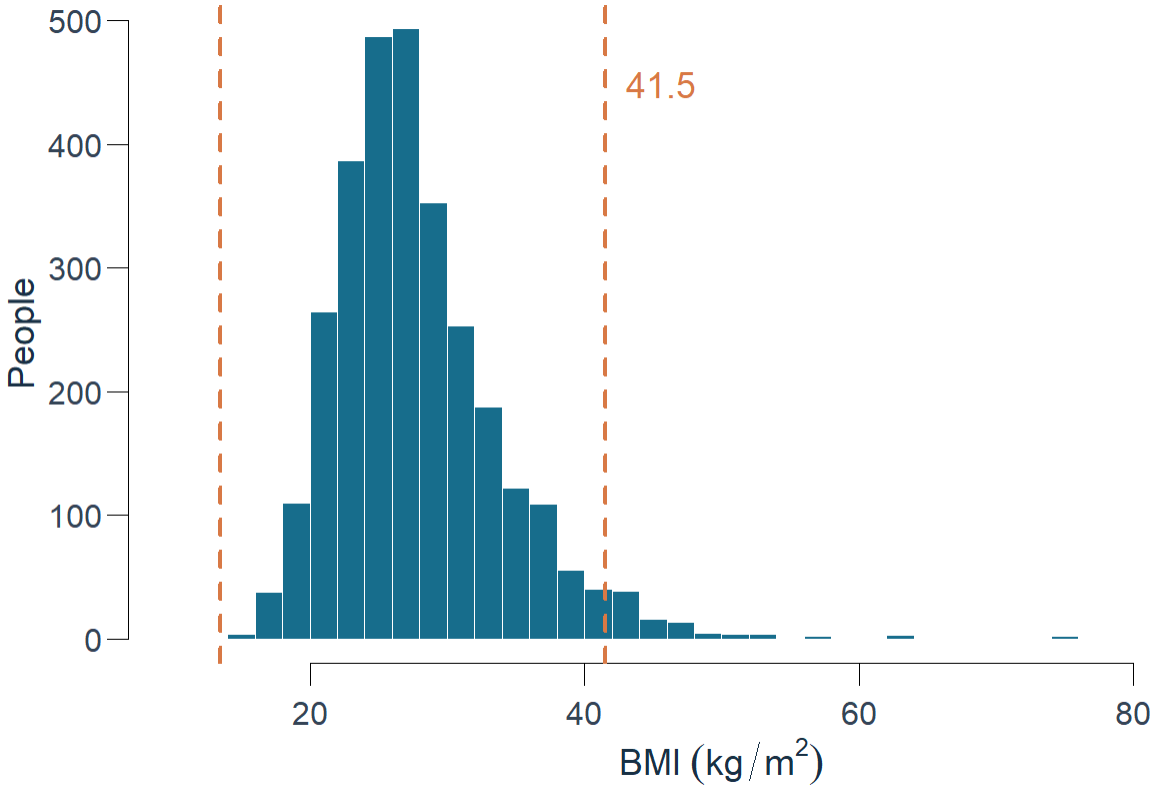
Import CSV → label categories → export clean data for the exercises

Readable labels and correct data types make every later analysis easier.

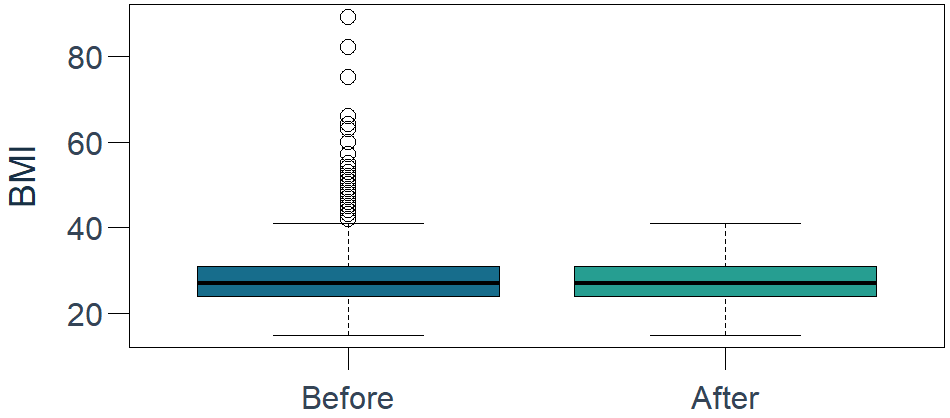
Outlier analysis and removal

Q1 = 24 • Q3 = 31 • IQR = 7 • Q1 − 1.5×IQR to Q3 + 1.5×IQR → 13.5 to 41.5

Detect unusual BMI values



Compare before and after



Measure	Before	After
People	3,000	2,886
Mean BMI	28.40	27.65
BMI variance	40.34	24.59
Positive status	15.63%	14.62%

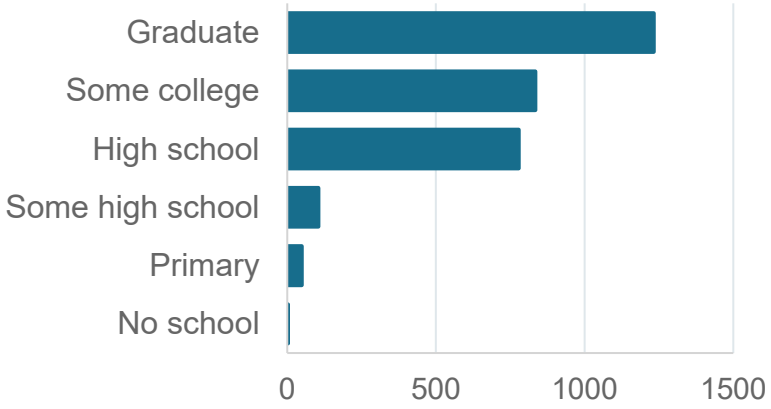
114 high BMI values (3.8%) were flagged. Removing them lowers the average and spread.

Summary statistics and graphs

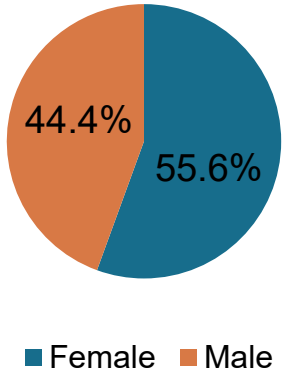
Mean = average • SD = standard deviation • Variance = squared spread

Variable	Mean	Median	SD	Variance
BMI	28.40	27	6.35	40.34
Mental-health days	3.26	0	7.39	54.67
Physical-health days	4.15	0	8.62	74.24

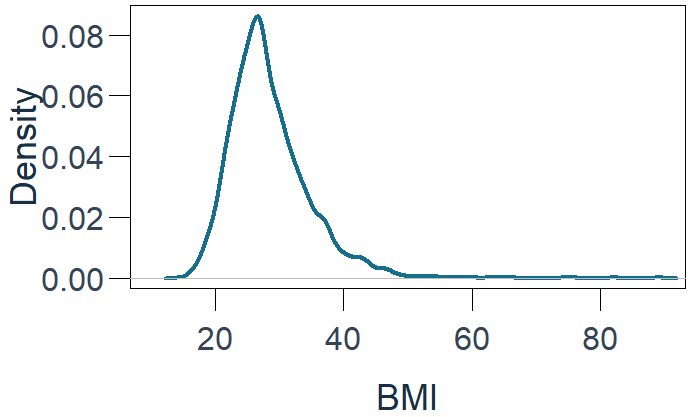
Education: group sizes



Sex: sample share



BMI: distribution shape

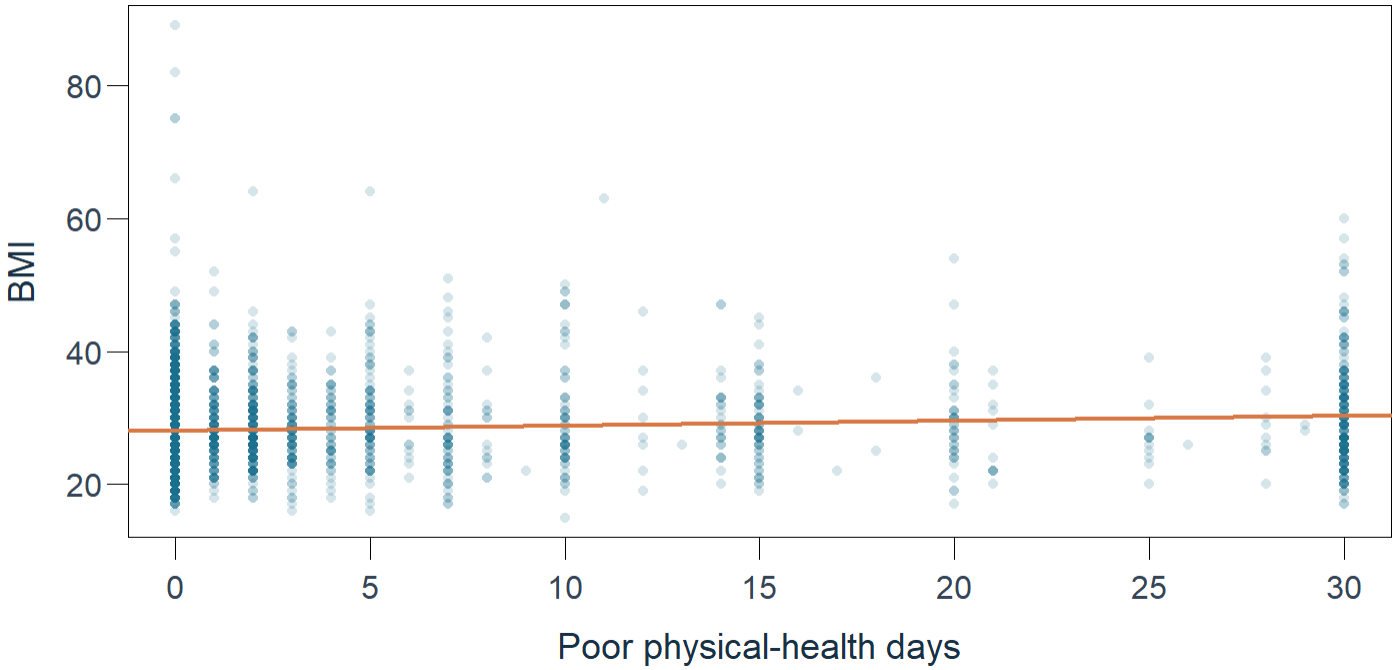


BMI has a long right tail. Most people report few poor-health days.

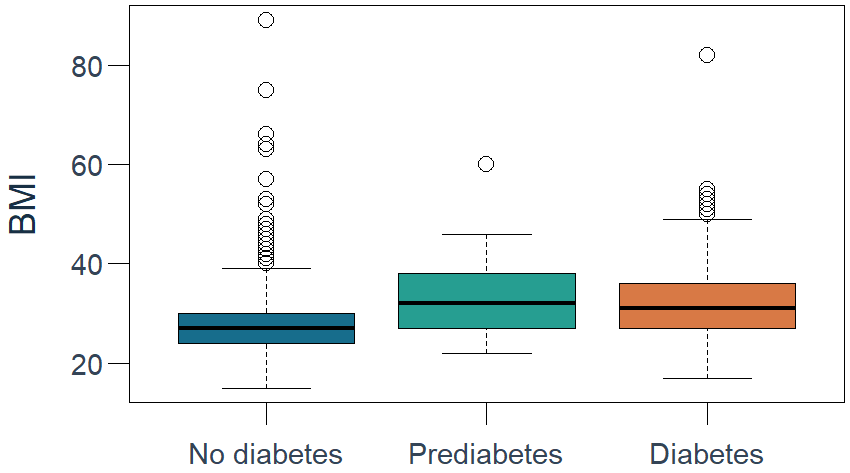
Correlation and simple regression

Predicted BMI = 28.09 + 0.0749 × poor physical-health days

Poor-health days and BMI



BMI by health group



Pearson r	Spearman r	Covariance	R^2
0.102	0.108	5.56	1.03%

Average BMI

No diabetes	Prediabetes	Diabetes
27.72	32.85	31.98

r = 0.102: weak positive relation. R^2 = 1.03%: very little BMI variation is explained.

Multiple regression and model checks

Predict BMI using age, activity, income, education and smoking. R^2 = variation explained.

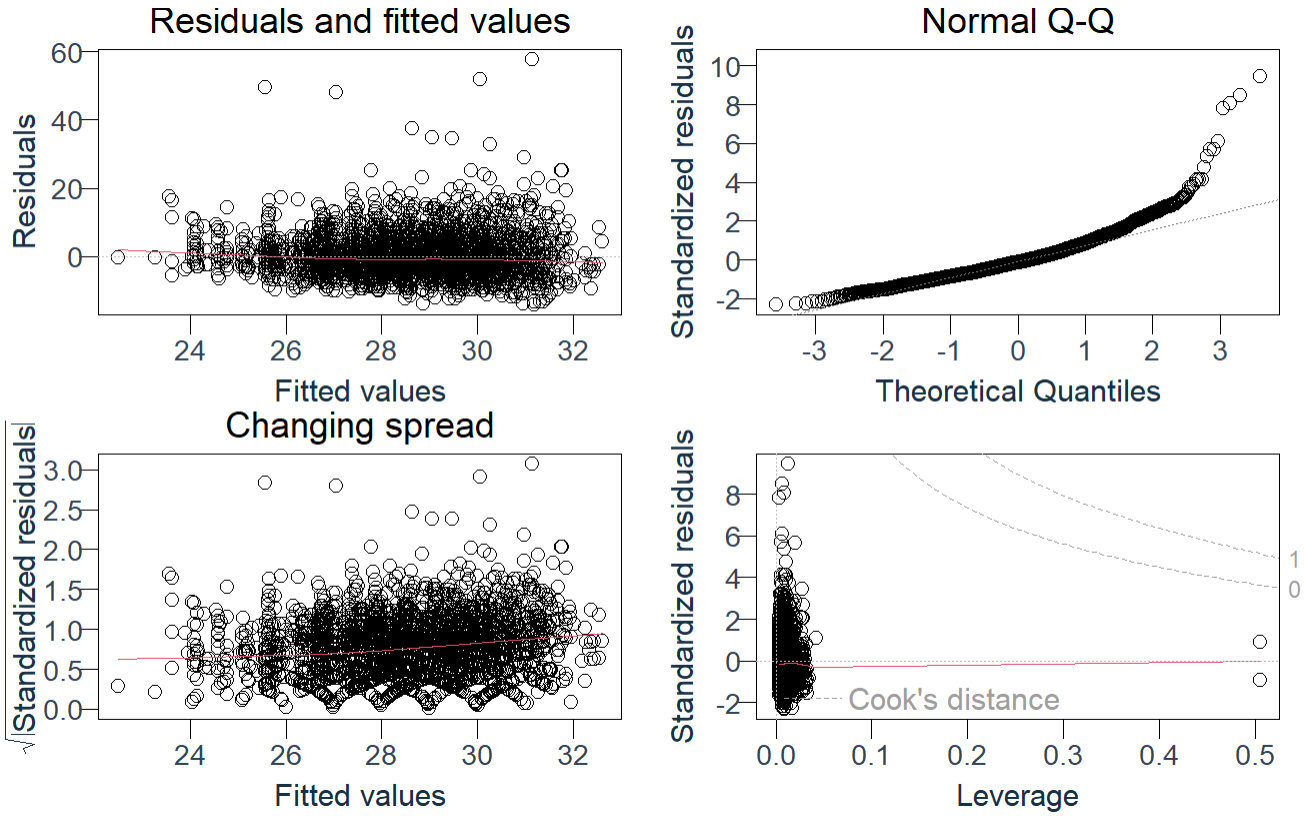
6.81%
R^2

5.99%
Adjusted R^2

Which predictors matter?

Predictor	F	p
Age	9.17	<0.001
Physical activity	13.30	<0.001
Income	4.43	<0.001
Education	8.03	<0.001
Smoking	0.07	0.786

Check the fitted model



Residuals show uneven spread and a long upper tail.

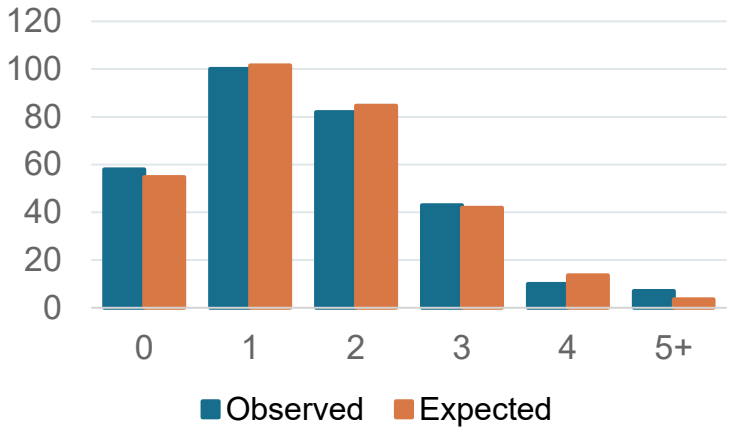
Age, activity, income and education relate to BMI. The model explains only a small share of variation.

Fitting probability distributions

Compare the observed pattern with the pattern each model expects.

Binomial

Positive cases in groups of 10.
p = 0.156, 300 groups.

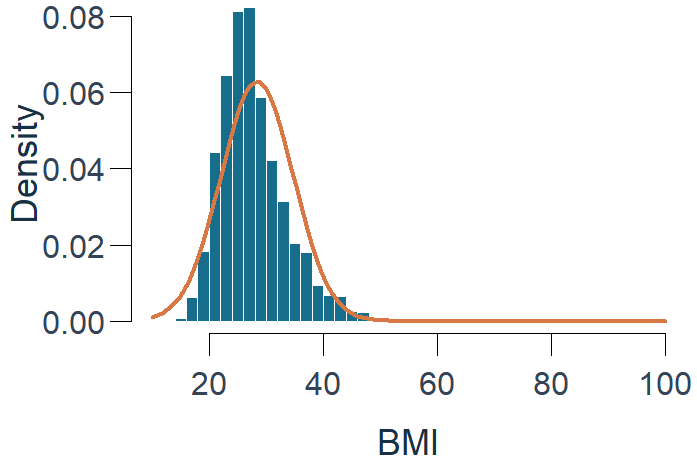


Test p = 0.962

Fits the grouped counts reasonably.

Normal

BMI: mean 28.40, SD 6.35.
The curve misses the long tail.

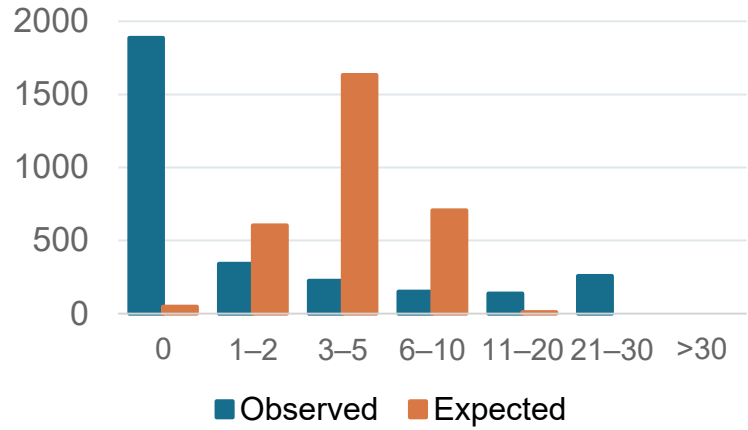


Test p < 0.001

BMI is not normally distributed.

Poisson

Physical-health days: mean 4.15,
variance 74.24. Many zeros.



Test p < 0.001

Variance is far above the mean.

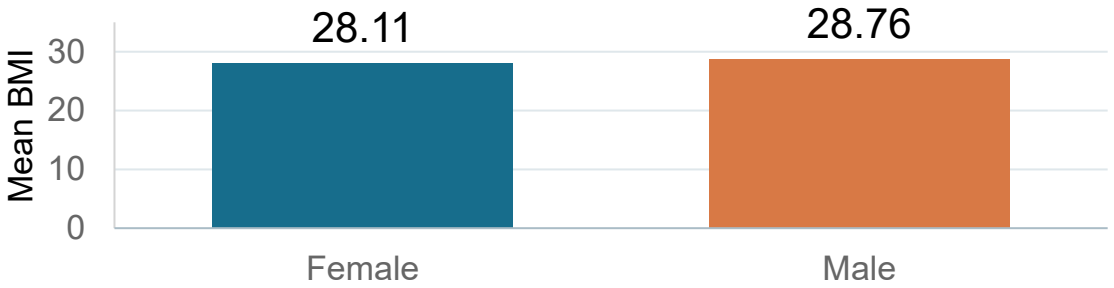
A useful analysis explains why a model fits or fails.

One- and two-sample hypothesis tests

Lab 7 compares a sample with a reference. Lab 8 compares two groups. We use α = 0.05.

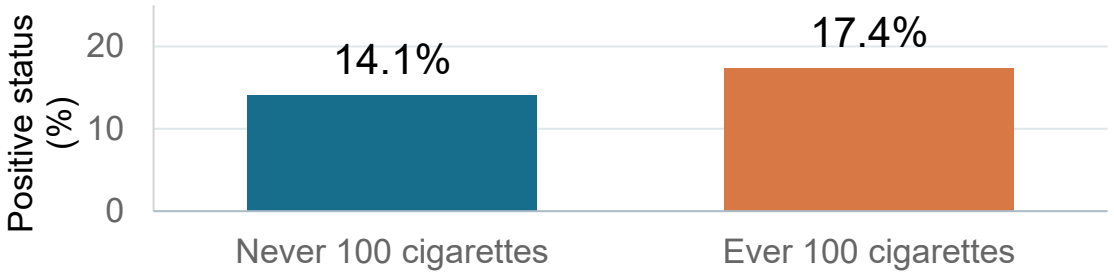
Question / comparison	Method	Result	p
Mean BMI vs 27	One-sample t	28.40 (above 27)	<0.001
Positive share vs 11%	One proportion	15.63% (above 11%)	<0.001
BMI: female − male	Welch t	-0.65 BMI units	0.004
Positive share: never / ever smoked	Two proportions	14.09% / 17.36%	0.016

BMI: compare female and male means



Female − male: -0.65 BMI units.
95% CI: -1.10 to -0.20.

Positive status: compare smoking groups



Never − ever: -3.27 percentage points.
95% CI: -5.95 to -0.59 points.

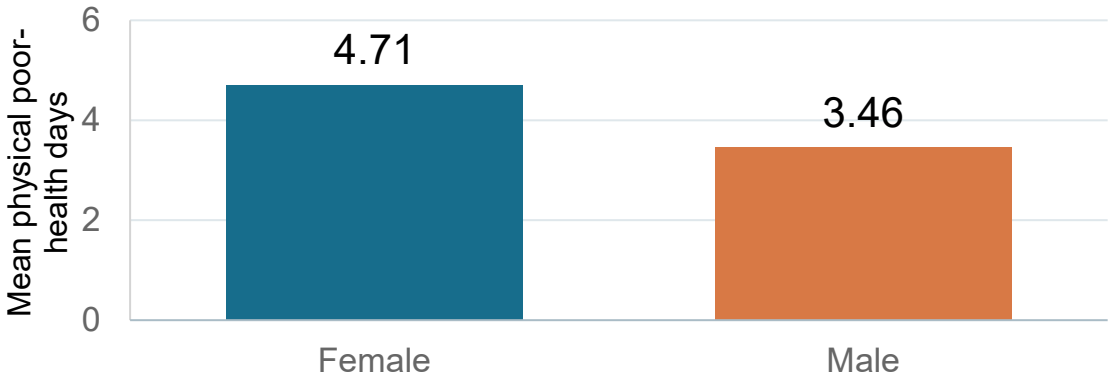
All four tests find differences at p < 0.05. References 27 and 11% are classroom assumptions.

Independent and paired t-tests

Independent tests compare different people. Paired tests compare the same person before and after.

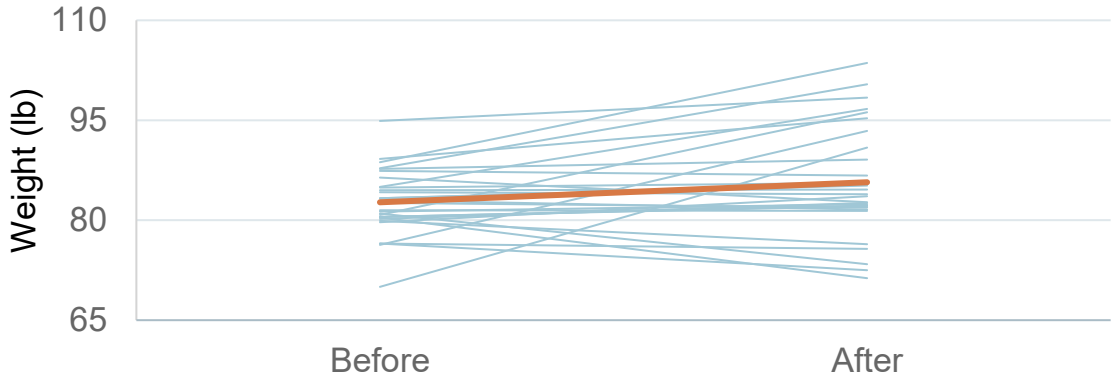
Independent: different respondents

BRFSS: 1,667 females and 1,333 males.
Compare physical poor-health days.



Paired: the same 29 CBT patients

MASS dataset: before/after weights (lb).
Blue lines: patients. Orange line: mean.



Independent Welch t-test	Result
Female − male	+1.25 days
95% CI	+0.63 to +1.86 days
p	<0.001

Paired t-test: after − before	Result
Mean change	+3.01 lb
95% CI	+0.23 to +5.79 lb
p	0.035

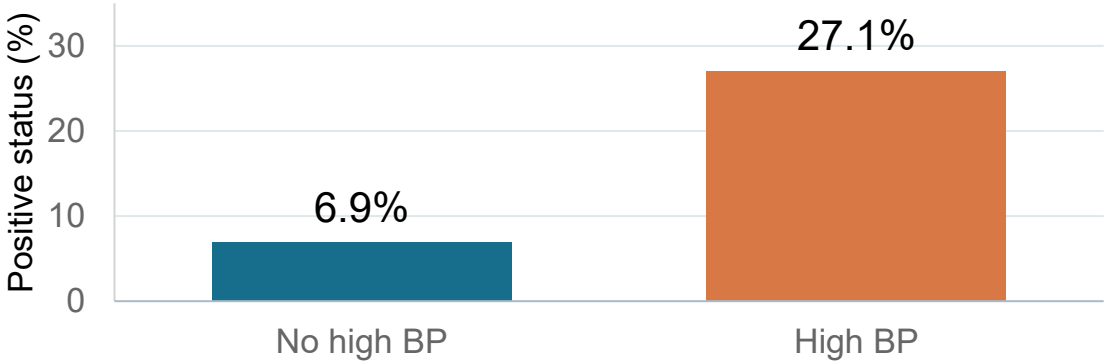
Females report more poor-health days on average. The paired average weight increases by 3.01 lb.

Chi-square: category patterns and relationships

Goodness of fit checks category counts. Independence tests check whether two categories are related.

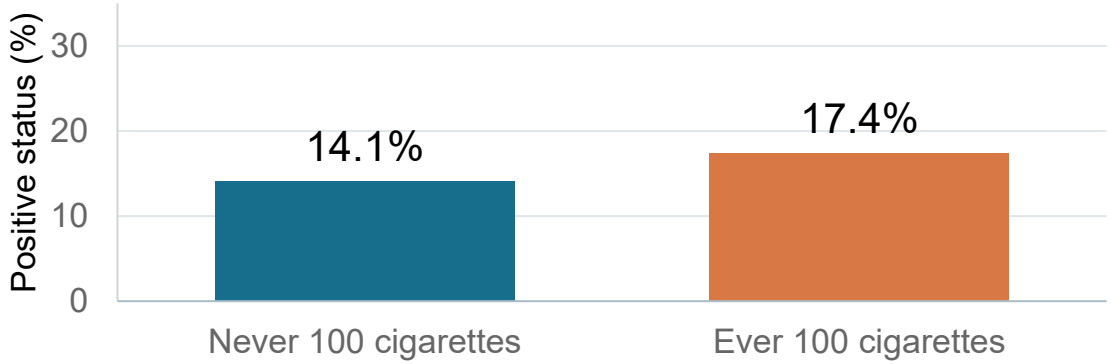
Test	χ^2	p
Age groups: equal frequency (GOF)	678.12	<0.001
Status × high BP	226.28	<0.001
Status × smoking	6.07	0.014

Blood pressure: larger difference



Association size: V = 0.275

Smoking: smaller difference



Association size: V = 0.045

Positive status is more closely related to high blood pressure than to ever-smoking in this sample.

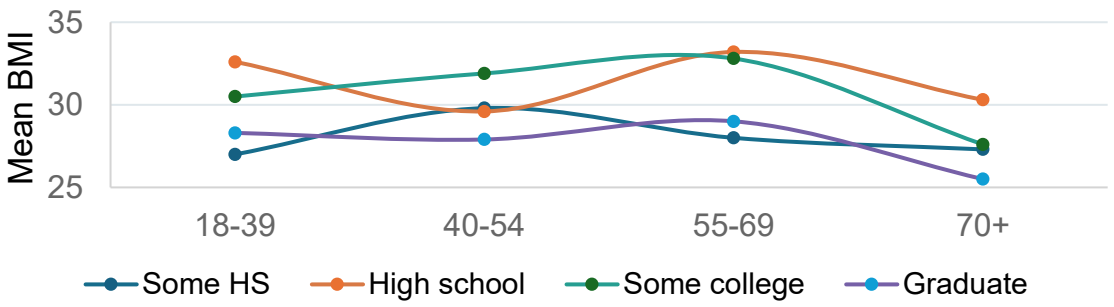
CRD, RBD and Latin Square using health records

BMI is the response. Education is the group factor. Age and income form blocks in matched subsets.

Analysis of education groups	Rows	F	p
CRD-form: one-way ANOVA	3000	14.41	<0.001
RBD layout: account for age	160	3.14	0.027
Latin Square layout: age + income	16	3.28	0.100

Full-sample checks: Welch p < 0.001. Holm: graduates differ from high school and some college (p < 0.001).

RBD layout: four age blocks



10 respondents in every age × education cell.

Latin Square: age and income blocks

Age	<20k	20–35k	35–75k	75k+
18–39	A	B	C	D
40–54	B	C	D	A
55–69	C	D	A	B
70+	D	A	B	C

A Some HS, B High school, C Some college, D Graduate.

All records come from BRFSS. Observational layouts: education was not randomly assigned.

Conclusion: findings and next steps

We described health patterns and applied every course method using documented data.

What we found

Higher average BMI in positive groups

27.72 without diabetes, 32.85 prediabetes, 31.98 diabetes.
Individual values still overlap.

Regression explains a small share

R^2: 1.03% (simple), 6.81% (multiple).
Most BMI variation remains unexplained.

Outliers and model checks matter

Removing 114 flags lowers mean BMI from 28.40 to 27.65.
Binomial fits reasonably. Normal and Poisson fit poorly.

The design changes the comparison

BP has a larger status association than smoking.
Blocks cut error by 44.8%. Wheat treatment C yields most.

Limits

2015, self-reported data.
Small groups add uncertainty.
Associations do not prove causes.
Low R^2 limits prediction.

Next steps

Use newer, better measurements.
Add useful BMI predictors.
Evaluate on unseen data.
Repeat the model checks.

All 11 labs are covered. The findings follow the data and their limitations.